Mahtab Foroozandeh

Interaction Designer & Aspiring HCI Researcher

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RESEARCH INTERESTS

I study how people think, decide and learn when interacting with AI and immersive technologies. My current research examines cognitive offloading to AI chatbots and its effect on memory, understanding and critical thinking, with the aim of designing interactions that support human thinking rather than replace it. I am also interested in XR and tangible interaction as ways to study embodied and emotional experience.

EDUCATION

Bachelor of Design (BDes), Interaction Design — Sep 2023 – May 2027 (expected)

Emily Carr University of Art + Design, Vancouver, BC

CGPA: 3.7 / 4.0 · Relevant coursework: Design Research Methods; Computer Science; 3D Modelling for Mixed Realities

RESEARCH EXPERIENCE

Graduation Research Project, Thinking with AI — Sep 2026 – Apr 2027

Emily Carr University · Supervisors: Dr. Manuhuia Barcham, Eugenia Bertulis

- Proposal: *User-Adjustable Levels of AI Involvement to Sustain Human Cognitive Engagement*, a "Thinking Dial" that lets users choose how much of the thinking AI does.
- Planned study: interviews and a one-week diary study (6–8 people), then a Wizard-of-Oz experiment (n = 32, 3 × 2 mixed design) measuring accuracy, over-reliance, retention, workload and trust.
- Completed TCPS 2: CORE research ethics training.

Design Researcher, Shumka Centre × PsychSafe — Aug 2026 – Present

Emily Carr University · Supervisor: Colby May

- Evaluating the accessibility and usability of a pre-launch workplace psychological-safety platform for users with varied mental-health literacy.
- Completed a heuristic evaluation; interviews with 4–5 pilot users and usability testing to follow.

Volunteer Research Assistant, Metacreation Lab for Creative AI — Jun 2026 – Aug 2026

Simon Fraser University, Surrey, BC

- Evaluated how well Autolume, the lab's generative-AI art tool, communicates what its models are doing, combining five human–AI interaction principles with Nielsen's heuristics and verifying claims against the source code.
- Identified 17 usability problems (2 major) and 5 prioritized recommendations, presented to the development team.

Design Researcher, Shumka Centre × Disposal Queen — Sep 2025 – Dec 2025

Emily Carr University · Supervisors: Colby May, Gill Barros

- Ran usability testing and competitive analysis of property managers' disposal workflows.
- As sole designer, redesigned the customer portal, which has since launched.

UX Researcher & Designer, HealthSync AI — Sep 2025 – Dec 2025

ISM Creatives (remote) · 3-person team

- Interviewed 3 people living with diabetes and analyzed Apple Health, Fitbit, Oura and MyChart; a key finding was that alert fatigue erodes trust.
- Independently designed the mobile and Apple Watch interfaces and delivered an interactive hi-fi prototype.

TECHNICAL EXPERIENCE

AI/ML Developer, Persustain (carbon-footprint app) — Jan 2026 – May 2026

ISM Creatives (remote)

- Built the AI pipeline that turns daily activities into carbon-emission estimates using rule-based emission factors and the Climatiq API.
- Used an LLM (GPT-4.1-mini) only for explanations; all numbers are computed in code, with JSON-schema outputs and audit logging to prevent hallucination.
- Stack: TypeScript, Node.js, Firebase Cloud Functions, Firestore, OpenAI API, Climatiq API.

SELECTED PROJECTS

The Endless Journey, Narrative Game on Forced Migration — Fall 2025

Core Design Studio III, Emily Carr · Solo designer & developer

- Turned interviews with two immigrants into game mechanics: an emotional-health meter that drains with isolation and NPCs that restore it through connection.
- Built Acts I–II in Unity 6.2 (HDRP, C#). Code: github.com/mahforooz/GameDesignProject

From Persepolis to Paris, WebVR Museum Experience — 2024

3D Modelling for Mixed Realities, Emily Carr · with Dorsa Nazarimanesh

- Modelled the Achaemenid Bull Capital, the Louvre courtyard and Persian carpets in Blender, and built the walkable experience in A-Frame for desktop and Meta Quest.
- Revised lighting, panel size and movement after headset testing with classmates.

Tangible & Physical Computing Projects — 2024 – 2026

Emily Carr · Arduino programming lead on three team projects, each tested with peers

- Calming armrest: grip pressure on a conductive sheet selects one of three calming audio tracks for nervous flyers.
- Responsive bus shelter: a floor pressure sensor lights the shelter for safer bus stops on dark, rainy Vancouver evenings.
- Interactive canvas: touch sensors trigger projected animations that layer onto a painting.

EXHIBITION

Persian Miniature, Max/MSP audiovisual work in which video speed follows the audio's tempo. End of Year Exhibition, Emily Carr University, Vancouver, 2024.

LEADERSHIP & SERVICE

Lead Mentor, FLUI (For the Love of UI) Hackathon, Interaction Design Department, Emily Carr University, March 2025.

DESIGN INDUSTRY EXPERIENCE

Visual Design Intern, Cuddles Learning Inc. (remote) — Aug 2024 – Jan 2025

Product Design Intern, SkillHat (remote) — Apr 2024 – Aug 2024

SKILLS

Programming: TypeScript/JavaScript, Python, C# (Unity), Arduino (C/C++), HTML/CSS
AI & data: OpenAI API (structured outputs), prompt design, Firebase/Firestore, data visualization
XR & physical: Unity, A-Frame (WebVR), Blender, Meta Quest, Arduino and sensors, Max/MSP
Research: Interviews, diary studies, participatory design, usability testing, heuristic evaluation, surveys, affinity mapping
Design tools: Figma, Adobe Creative Suite, Miro, Framer, Spline
Languages: English (fluent), Persian (native), French (A2)
Training: Google UX Research; CalArts UI/UX Design; CalArts Graphic Design